Certified representation and partial-slope pruning

Source-only mathematical designs; both independently reviewed

Euler representation correction

Full-cube wire aliases preserve projected solutions. No range CNF needed.

Checked linear DAG nodes avoid flat multiplier expansion.

Target-independent membership: no extra pruning after factor-base filtering.

Target-dependent trace filter: finite constant-B k2 chart

$q^2 = a + 2t + z$, $0 \le z \le 2B-2$ ⟶ complete necessary slope set T

Verify every offset ⟶ construct LSB trie ⟶ reject missing partial prefixes

At d = s+3: ≥4B+2 of 8B prefixes empty, if s+3 ≤ k−1.
Source-only derivation · setup and verification charged · no execution

Euler aliases remove an unnecessary domain prerequisite

Full k-bit wires map surjectively to the prime field modulo p. Every bit solution gives the intended field computation; every valid projected pair has a canonical bit extension. Thus aliases preserve the projected set. The existing adapter accepts the full auxiliary cube and deduplicates projected models. Per-wire canonical ranges are optional, correcting the earlier dependency while preserving its frozen record.

For fixed invalid x, checked linear nodes obey $E_j = G_j + Z_b E_a + c_a E_b$. The final identity $1 = A/2 - (W-2)E_{W/2}$ gives a degree-two PC derivation with retained circuit variables. A specialized checker can verify these local identities in $O(Kk^2)$ field-term operations without expanding ancestor multipliers. Parsing, masks, bit arithmetic, authentication, storage, coverage and witness costs remain charged. A new authenticated arm/wrapper and DAG verifier are still required.

Membership depends on the curve and coordinate, not the target. After factor-base membership filtering it rejects no additional coordinate. This is a representation/proof design, not new relation-dependent pruning.

A target-dependent trace filter

In the exact finite-target constant-B k2 chart, the source rows give $A0 = b - aq$, $u1 = a - q^2$ and $u0 = \text{alpha} + a^2 + a\,q^2 - 2bq$. The remaining norm row is the target curve equation. For interval roots t+d1 and t+d2, counted with multiplicity, Vieta implies

$$q^2 = a+2t+z, \text{ where } 0 \le z \le 2B-2 \text{ and } B = 2\text{^}s.$$

The necessary slope set T has at most min(p,4B−2) elements. A certificate covers every offset with zero, both checked square roots, or an independently checked nonsquare Euler value. The verifier derives T and its LSB trie. Missing paths certify rejection before all slope bits are assigned.

At depth $d \le \text{floor}(\log_2 p)$, at least $\max(0, 2\text{^}d - (4B-2))$ canonical-completable prefixes are empty. If $s+3 \le k-1$, depth s+3 rejects at least 4B+2 of 8B prefixes, leaving at least one slope bit free. No distribution premise is needed. Empty depth-d cylinders need not be distinct minimal clauses; a free bit need not give two canonical completions.

Boundaries and next step

This applies to transformed q-first k2 modes, not the original A0-first projection or other chart strata. A semantic certificate verifier must bind the exact rows, interval theorem and complete table; the old flat-PC checker does not accept a trie automatically. Remaining candidates still need all chart/sign/witness checks. Setup and root extraction are charged and must be compared against direct O(B)-scale interval point enumeration.

No candidate execution, implementation, new slot or retry occurred. The 106+6/five-lead scope remains. Publish the reviewed designs, then specify the new checker interfaces; no runtime or degree-of-regularity claim follows.

Sources: design-v2.md and independent-review.md (alias/DAG); ../prime-k2-trace-prefix-20261108/design-v1.md, prefix-count-addendum-v1.md and independent-review.md (trace). Exact source hashes are in each source-bindings.json. Adapter full-cube/projection semantics; transform.py literal k2 rows and projection; composition.py interval idempotents/nilpotence. Prior frozen documents remain unchanged.

Literature: MIT18.783 (2021), Lecture3, sections3.10/3.13: randomized root finding and verifiable outputs; root cost remains explicit. Eurocrypt2014 PDF fetch returned403 as reported by root; no full-text review or novelty clearance.